

# Cafeteria Order Data Analysis

## Hi, I'm Sakshi Adhikari

For this assignment, I was given a cafeteria order dataset and asked to perform data cleaning, exploratory data analysis, identify useful business insights, and build a simple model to forecast the next 7 days of orders for a selected branch.

At first, the assignment looked straightforward: import the SQL database, clean the data, analyze it, and build a forecast. But once I started working with the actual dataset, I realized that the biggest part of the assignment was not just analysis. It was figuring out **how to work with a large dataset efficiently and make sure that the conclusions I presented were actually reliable.**

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## 1. Starting Out – Importing the Dataset

The first challenge appeared before I could even start the analysis.

The dataset was provided as a MySQL SQL file containing a very large number of records. I created a database called `cafeteria_db` and started importing the data using MySQL. The import was extremely slow because of the size of the `orders` table. At one point, MySQL appeared to hang while inserting records.

Since the assignment deadline was approaching, I realized that continuously waiting for the complete import or running heavy queries was not going to be an efficient approach.

So instead of repeatedly trying to process the entire table, I changed my approach. I used MySQL to access the database and created **targeted extracts** from the `orders` table. I extracted three separate samples of 50,000 records each, giving me a total working dataset of **150,000 orders**.

The extracts covered different portions of the original data, including records from:

- April 2024
- July–August 2024
- December 2024

This allowed me to move the analysis into Python and work with the data much more efficiently

## 2. Exploring the Database

Before analyzing the data, I explored the available tables to understand what information was present.

The database contained several order-, dish-, category-, and cart-related tables. I inspected tables such as `orders`, `carts`, `dishes`, and `category_has_dishes`. One thing I learned during this step was that having multiple tables does not automatically mean that they should all be joined. For example, the `carts` table contained useful item-level information such as `dish_id`, `quantity`, `dish_price`, and `total_price`, but it did not contain a validated direct `order_id` relationship that I could safely use to connect every cart record to an order. Rather than creating an unverified join and potentially producing incorrect item-level results, I decided to focus my main analysis on the `orders` table, where the transaction-level relationships were clear.

## 3. Moving the Analysis to Python

After extracting the data, I loaded the three datasets into Python using Pandas and combined them into one working DataFrame.

The final working dataset contained:

**150,000 orders**

The data covered:

**April 1, 2024 to December 4, 2024**

and included four branches:

**1, 2, 4 and 9**

I converted the date fields into proper datetime format and created additional columns such as:

- Date
- Hour
- Day of the week
- Month

I also converted the financial columns into numeric values so that revenue and average order value could be calculated correctly.

There were missing values in some fields. For example, `discount_amount` had many missing values, so I treated missing discounts as zero for the analysis. Missing payment-method values were grouped as **Unknown** instead of removing those transactions.

After cleaning the working columns, I was able to proceed with the exploratory analysis.

## 4. What Did the Data Tell Me?

### Branch Performance

One of the first things I wanted to understand was how the branches were performing.

| Branch   | Orders | Revenue       | Avg. Order Value |
|----------|--------|---------------|------------------|
| Branch 2 | 69,203 | ₹41,42,620.83 | ₹59.86           |
| Branch 1 | 60,505 | ₹41,41,897.41 | ₹68.46           |
| Branch 4 | 10,265 | ₹5,54,565.00  | ₹54.02           |
| Branch 9 | 10,027 | ₹5,72,867.35  | ₹57.13           |

Branches 1 and 2 clearly represented the majority of the sampled activity.

Together, they accounted for approximately **86.5% of the sampled orders** and around **88% of the sampled revenue**.

Another interesting observation was that Branch 1 and Branch 2 generated almost the same sampled revenue even though Branch 2 had more orders.

This was because Branch 1 had a higher average order value:

**Branch 1 – ₹68.46**

**Branch 2 – ₹59.86**

This showed me that looking only at the number of orders does not always tell the complete business story.

## 5. When Are Orders Happening?

I also analyzed the orders by hour to understand the busiest periods.

The highest observed order volumes were:

| Hour | Orders |
|------|--------|
|------|--------|

|       |        |
|-------|--------|
| 19:00 | 18,329 |
|-------|--------|

|       |        |
|-------|--------|
| 23:00 | 15,816 |
|-------|--------|

|       |        |
|-------|--------|
| 22:00 | 15,411 |
|-------|--------|

|       |        |
|-------|--------|
| 18:00 | 13,961 |
|-------|--------|

The strongest concentration in my sample was therefore during the **18:00–23:00 period**.

The 19:00 hour had the highest observed order volume, with **18,329 orders**.

I also calculated average order value by hour. The highest average order values occurred at different hours from the highest order volumes.

For example, the average order value around 18:00 was approximately **₹79.08**, while the 19:00 average was approximately **₹78.88**.

This was interesting because it showed that the busiest period and the highest-value period do not necessarily have to be exactly the same.

## 6. How Are Customers Ordering?

The dataset also contained information about how the orders were placed.

| Order Through | Orders | Revenue       | Avg. Order Value |
|---------------|--------|---------------|------------------|
| Mobile App    | 91,453 | ₹51,81,203.59 | ₹56.65           |
| SOK           | 32,512 | ₹25,74,953.00 | ₹79.20           |
| POS           | 26,035 | ₹16,55,794.00 | ₹63.60           |

The **mobile app** accounted for about **61% of the sampled orders**, making it the largest ordering channel.

However, SOK had the highest average order value at approximately **₹79.20**.

This gave me another useful perspective: a channel can generate a large number of transactions without necessarily having the highest value per transaction.

## 7. Payment Method Analysis

I also looked at payment methods. After standardizing variations such as `paytm`, `Paytm`, `upi`, `UPI`, and similar values, I obtained:

| Payment Method | Orders | Order Share |
|----------------|--------|-------------|
| Paytm          | 61,334 | 40.89%      |
| UPI            | 30,224 | 20.15%      |
| CCA            | 15,640 | 10.43%      |
| Cash           | 11,074 | 7.38%       |
| QR             | 10,964 | 7.31%       |
| Razorpay       | 10,777 | 7.18%       |
| Card           | 6,285  | 4.19%       |
| Unknown        | 3,702  | 2.47%       |

Paytm and UPI together represented approximately **61% of sampled orders**.

This suggests that digital payment methods formed a major part of the transaction activity in the dataset.

## 8. A Challenge I Found in the Data

While exploring the dataset, I also noticed something important.

All 150,000 extracted records had the same value for `order_status`.

Because of this, I could not meaningfully compare completed, cancelled, or other order statuses using this particular extract.

I decided not to create a misleading chart just to have another visualization in the report. This was another useful lesson from the assignment: sometimes the correct analytical decision is to say that a particular comparison cannot be reliably made from the available data.

## 9. Key Insights I Took From the Analysis

After completing the analysis, these were the main observations I found:

### 1. Branches 1 and 2 dominate the sampled activity

They accounted for approximately **86.5% of orders** and **88% of sampled revenue**.

### 2. Branch 1 has a higher average order value

Branch 1 generated almost the same sampled revenue as Branch 2 despite having fewer orders.

### 3. Evening is the strongest observed ordering period

The largest order volumes were concentrated between approximately **18:00 and 23:00**, with 19:00 being the highest observed hour.

### 4. Mobile ordering is significant

The mobile app represented approximately **61% of sampled orders**.

### 5. SOK has the highest average order value

Although SOK had fewer orders than the mobile app, its average order value was approximately **₹79.20**.

### 6. Digital payments dominate

Paytm and UPI together accounted for approximately **61% of sampled orders**.

### 7. Data quality and structure matter

Missing values, inconsistent payment-method names, non-continuous extracted periods, and unclear relationships between some tables all affected how the analysis could responsibly be performed.

## 10. What I Learned From This Assignment

When I started this assignment, I thought the main challenge would be writing Python code and creating charts. Instead, I learned that data analysis begins much earlier than that.

The first challenge was simply getting a very large dataset into a form that I could work with. I had to deal with slow SQL operations and MySQL becoming unresponsive during large processing tasks. Rather than repeatedly running expensive queries, I changed my approach and created smaller, targeted extracts.

I also learned that database exploration is important before writing joins. Just because two tables contain related-looking fields does not mean that they can safely be joined.

Most importantly, I learned that **data analysis is not about forcing an answer out of the data. It is about understanding what the data can actually support.**

For the forecasting part, I could have produced a 7-day forecast from the extracted data, but because the sample was not a continuous daily time series. That decision made the project more realistic for me.

## 11. Final Reflection

This assignment gave me experience with much more than Pandas and visualization.

I worked through a real-world type of problem where the dataset was large, the database was not immediately easy to process, some fields required cleaning, and not every analysis could be performed exactly as originally planned.

My final analysis focused on the parts of the data that I could validate properly.

The main files I prepared for the project include:

- Python analysis notebook
- SQL queries used for data extraction
- Key insights and recommendations
- Final project report

The biggest takeaway for me was simple:

**Good data analysis is not about using the most complicated model. It is about making sure that the data, method, and conclusion actually match each other.**